

Integration of Internet-of-Things as sustainable smart farming technology for the rearing of black soldier fly to mitigate food waste

Josiah Cheng Foong Van^a, Pei En Tham^a, Hooi Ren Lim^a, Kuan Shiong Khoo^b,
Jo-Shu Chang^{c,d,e,*}, Pau Loke Show^{a,*}

^a Department of Chemical and Environmental Engineering, Faculty of Science and Engineering, University of Nottingham Malaysia, Jalan Broga, Semenyih, Selangor Darul Ehsan 43500, Malaysia

^b Faculty of Applied Sciences, UCSI University Campus No. 1, Jalan Menara Gading, UCSI Heights, Cheras, Kuala Lumpur 56000, Malaysia

^c Department of Chemical Engineering, National Cheng Kung University, Tainan 701, Taiwan

^d Department of Chemical and Materials Engineering, Tunghai University, Taichung 407, Taiwan

^e Research Center for Smart Sustainable Circular Economy, Tunghai University, Taichung 407, Taiwan

ARTICLE INFO

Article History:

Received 25 November 2021

Revised 12 January 2022

Accepted 23 January 2022

Available online 10 February 2022

Keywords:

Black soldier fly

Internet-of-Things (IoT)

Food waste

Sensor

Smart farming

ABSTRACT

Background: Throughout the world, food wastage issues continue to plague almost every country. Multiple ideas and solutions have been conceived and are continuously being tested by scientists and government bodies to mitigate food waste management issues. Black Soldier Fly (BSF) rearing is an up-and-coming commodity because of its versatility and multi-function purposes in various fields, such as food waste management, animal feed industry and bioactive compounds industry.

Methods: This work looks at setting up an automated smart farming system to rear BSF, with the help of implementing the Internet-of-Things (IoT) into the monitoring system. It also entails a guide on a possible design of a home-based Black-Soldier-Fly smart farm, where the Internet-of-Things components such as sensors, relays, and mobile applications are showcased. Finally, the prospects and challenges that arise with Black-Soldier-Fly smart farming can be identified and discussed.

Significant findings: Important growth factors such as temperature, light and pH can be monitored remotely by Internet-of-Things technology. Through IoT implementation, the farm can be remotely controlled and growth parameters can be adjusted with ease. Hence, this would lead to the efficient production of BSF larvae for processing food waste or conversion to bioactive compounds.

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1. Introduction

By 2050, it is projected that 9 billion people will inhabit the planet [1]. This means that the global food demand will rise, which requires greater development in the agricultural field. However recent studies of current trends in food availability and demand show that the yield produced will not be enough to meet the required global food demand [2]. Also, a greater drive in agriculture would contribute to climate change, different kinds of pollution and food waste issues. Food waste is becoming a prevalent issue in the world due to its implications in economic and environmental scenarios, as well as it is related to climate change [3,4]. Hence, there is an urgency in searching for suitable methods to tackle food waste issues that are environmental-friendly and effective.

Conventional methods to mitigate food waste such as landfills are not ideal as they create execrable odors, leachate and landfill gases which are sources for global warming, as well as the fact that eventually, landfills around the world will reach maximum capacity [5,6]. It has been found that an insect known as the black soldier fly (BSF), can reduce certain food waste types such as kitchen waste, fruits and vegetables, and poultry feed [7]. They are also capable of valorizing this food waste by converting them into insect biomass, which in turn can be used as feed for livestock [8]. More recently, biofertilizers have also been converted, leading to a more sustainable agricultural approach [9]. These studies show that the black soldier fly has great potential to be a solution to global food waste issues.

Black soldier fly (BSF), *Hermetia illucens*, is an insect of the order Diptera and part of the Stratiomyidae family. The fly has been known to be native to the North American region but is now found throughout the world [10]. However, there is an absence of its possible habitation in the northwestern part of Europe due to its low cold hardiness [11]. The BSF and its larvae are unique as compared to similar fly species, they are not considered pests or vectors [12] as they do

* Corresponding authors.

E-mail addresses: changjs@mail.ncku.edu.tw (J.-S. Chang), Show@nottingham.edu.my (P.L. Show).

not possess mouthpieces or stingers which can cause harm to humans [13]. Due to this, BSF has been widely utilized in many fields such as aquaculture, waste management, and the animal feed industry [14]. It is noted that the key interest of the BSF life cycle is the larval stage, as the larvae carry many benefits such as the roles played in the aforementioned fields. The BSF larvae also have a high content of protein, lipids and bioactive compounds, which have great potential to be extracted for human and animal well-being [15,16].

As the rearing of BSF larvae carries great prospects, many farmers are turning to both small and large-scale production and breeding of the larvae. The rearing process usually involves a lot of manual labor and manpower, which can be seen as a tedious and troublesome process. Site location, financial constraints and farm design are factors that also need to be taken into account when starting a BSF farm [17]. Large-scale designs typically use large concrete basins with one side having a sloping ramp leading to a rain gutter to collect the larvae [18]. BSF farms also require stringent optimum conditions to maximize the production of BSF larvae. As we are moving forward to an era of digitalization, it is important to look at innovations and technology to solve the difficulties faced in manual processes [19].

The world of human beings and technological devices are connected through the Internet, which allows the ease of communication and transfer of information. Hence, the innovations and technology mentioned above typically revolve around this relationship, which leads to the concept of the Internet-of-Things (IoT). It is a relatively new concept that is extremely relevant in today's world. It has a great potential in being implemented into the rearing of BSF larvae, which has been traditionally a laboring industry. The conjunction of the principles of IoT with traditional farming has led to the concept of smart farming. When mentioning new technologies in the BSF field, it does not necessarily mean that all components would need to be connected to the Internet. Recently, Jeff Tomberlin and his research team have developed a BSF 'bullet' technology, which is a unit that helps to maintain BSF larvae in a juvenile state for longer periods before being used for any intended purpose [20]. Regardless, any advancements in innovation and technology in the BSF field would tremendously be beneficial to the industry.

In recent years, researchers from India have set up an electrically controlled artificial system to manage organic waste through the

usage of BSF with IoT monitoring. The main purpose of this research was to design a system with minimum human interaction which allowed remote IoT monitoring from any part of the world [21]. Besides that, in Sarawak, Malaysia, researchers were successful in setting up a BSF farm using IoT technology. This was done to show that insect rearing with IoT monitoring was capable of being carried out in rural areas, where conventionally manual farming was carried out [22]. These examples (discussed in more detail in Section 3.4) show the potential of IoT being used in BSF rearing.

The main objective of this review is to investigate necessary steps needed to be taken to set up a BSF smart farm from the comforts of one's home and to be able to monitor the insects' growth through IoT means. This review primarily focuses on a small-scale approach as compared to other reviews, which allows researchers to have a better understanding of the growth criteria, before scaling to a larger industrial scale. To understand the process of setting up a BSF smart farm, there needs to be a clear understanding of the life cycle and growth parameters to set up the idealized and optimized conditions for successful growth. By setting up a BSF smart farm with IoT implementation at home, one can experience rearing insects on a small-scale, thus this experience and knowledge can be transferred when planning to carry out a smart farm on large scale. Besides that, through this review, researchers can set up BSF smart farms in research laboratories to carry out observations or extraction to extract the vital bioactive compounds contained within the insects, or experiment with food waste management approaches. Through small-scale smart farming, growth parameters can be adjusted to one's requirements and allow for ease of obtaining experimental subjects to carry out scientific research.

2. The farming process in BSF

2.1. Rearing of BSF

BSF has a life cycle of about 45 days, which can be split into four distinct stages: egg, larva, pupa, and adult stages [23]. Fig. 1 below gives a visual representation of the BSF life cycle. BSF begin the life cycle as eggs for about 4 days, then they hatch into larvae. The larvae are omnivorous and voracious, hence they will consume food waste

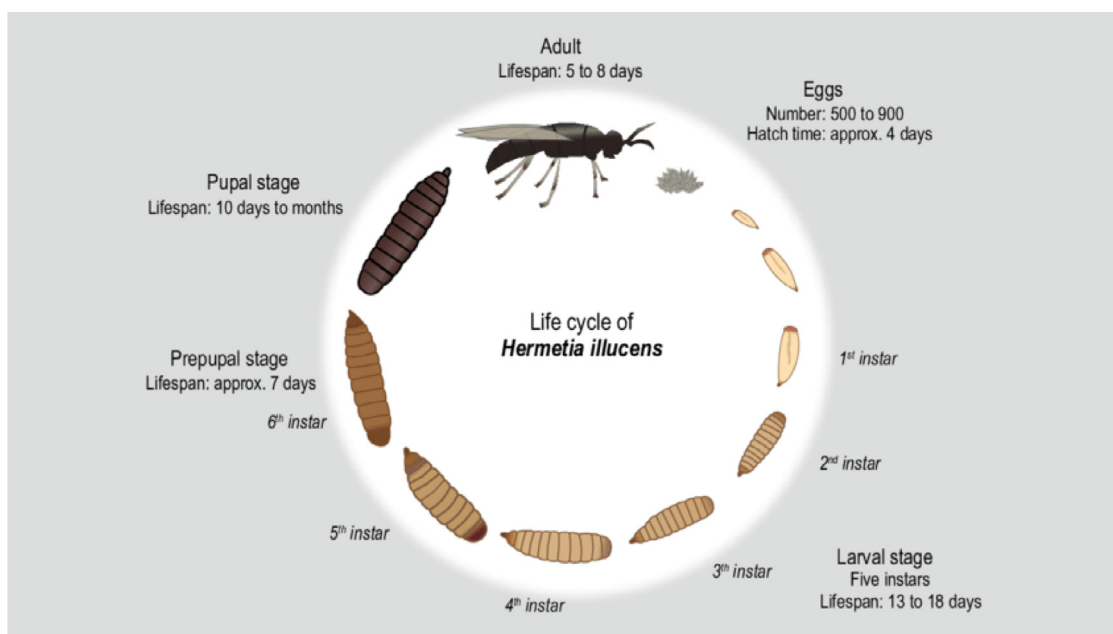


Fig. 1. The life cycle of the black soldier fly (Permission obtained from [28] that was modified from [29]).

with great efficiency, causing them to grow rapidly and leave behind a nutrient-rich fertilizer known as frass [24,25]. After 2–3 weeks, the larvae begin turning to pupae and become more inactive and immobile. They will also look for a dry space to undergo metamorphosis into flies (adults) [23]. Adult flies have a lifespan of about a week before they die [26], but the lifespan can vary depending on the conditions. During the adult stage, the flies do not have mouthpieces and do not feed on perishables. However, a study in Japan has shown that when sugar and water are placed near newly emerged adults, their longevity increases by 3 times in males and 2 times in females, as compared to providing water alone [27]. This goes against the consensus that BSF adults need not feed on food to survive for a greater lifespan. Hence, more detailed research needs to be carried out on the need for the consumption of food for BSF to increase adult longevity.

2.2. Growth parameters

2.2.1. Temperature

BSF tend to be very sensitive to changes in their surroundings, hence meticulous care needs to be given when rearing them. The development of BSF larvae can be affected by many growth parameters such as temperature, diet, humidity, and light intensity. These factors need to be in optimum condition to ensure proper growth. It is found that the ideal temperature for BSF larvae to successfully nurture is in between the range of 24–30 °C [18,26,30]. Temperatures below and above the minimum and maximum threshold of 15 °C and 40 °C, respectively are shown to be unfavourable for the growth of BSF in all its life stages, causing complete mortality in the species [18,31]. Temperatures that are too cold can cause the BSF larvae to have slowed metabolism, inactiveness hence unable to develop properly. On the other hand, temperatures that are too high cause the BSF larvae to search for cooler places to rest instead of feeding [26]. BSF has also been shown to be greatly affected by temperature differences. BSF larvae and adults averaged survivance of 74–97% when reared at 27–30 °C, but attained a survivance of only 0.1% at 36 °C [32]. It is also found that the ideal mating and fecundity temperature is 28 °C [33]. Hence, the *Hermetia illucens* can be said to be a delicate species, as it has very specific temperature requirements to ensure optimum growth.

2.2.2. Diets

The dietary regime of the BSF also plays a big role in the development of the insect. amongst its life stages, it mostly consumes food during the larval stage. BSF larva can consume a large amount of food and grow many times bigger than their original size [34]. It has been noted and highlighted that nutritional components such as carbohydrates and proteins play an important role in the growth of insect larvae [35–37]. Agricultural students from a Dutch agricultural college have found the optimum diet formulation for BSF larvae. The optimum diet consists of 66.96% protein, 8.93% starch, and 24.11% roughage to help with the digestibility of the meal [38]. Experiments were done and showed that the best protein source were ethanol by-products and corn steep, which resulted in the greatest weight gain [38]. For starch sources, mashed potatoes, wheat starch and potato pulp were options that performed the best. Finally, corn gluten meal was a good choice for roughage to give the diet structure. Hence, a food waste consisting of components from each of the three criteria would result in the optimum diet for BSF larvae. Moisture on the feed also plays a big role in nurturing the BSF larvae. A feed containing a moisture content of 40% causes the larvae to unsuccessfully develop, while a feed of moisture content of 70% helped to develop the larvae the fastest and increased their survival rate [39]. In terms of selecting the protein source for BSF larvae consumption, it is found that a pork diet is inferior to a beef diet and grain-based diet, as it increased the larval development time by about 23% and 140% more degree hours

respectively [40]. BSF larvae fed on fish waste *Sardinella aurita* also proved to have a high protein content of 78.8%, while food waste consisting of fruits and vegetables only yielded a protein content of 12.9% [41]. Besides that, from personal experiences in rearing BSF larvae, feeding a diet containing a lot of vegetable and food scrap waste, will cause a large amount of ammonia to be produced which can prove to be toxic in high amounts and very uncomfortable for people in the vicinity. A study also showed that BSF larvae focused on vegetable composting increased the concentration of ammonia in frass by about five to six-fold as compared to conventional means [42]. There is a prospect of adding coffee grounds into the dietary needs of BSF larvae, as coffee grounds are high in protein and caffeine. In other studies, some insects that were fed coffee grounds showed greater metabolism and locomotor activities [18,43]. However, more research is needed to be carried out on the effects of caffeine on the growth of *Hermetia illucens*.

2.2.3. Light

Light intensity is also a factor in the life cycle of the BSF, particularly during the mating and fertilization process [33,44,45]. Both natural lighting (i.e. sunlight) and artificial lighting (quartz iodine lamp, LED lamps) promoted mating [33], but LED lamps promoted BSF egg fertilization, whereas sunlight encouraged improved hatchability and fertility [27], amongst all the artificial light sources available on the market, BSF LED lamps (specially designed lamps based on BSF adult visual spectral sensitivity) provided to be the most energy-efficient light source and also produced the highest mating success rate based on inseminated females [46].

2.2.4. pH

On the other hand, the effects of pH on the development of BSF larvae were tested and experimented with. On one occasion, it was found that even though different pH environments ranging from values of 4.0–9.5 were tested, however, there were no significant differences in final mass between the tested samples of BSF larvae [47]. Another study showed that lower pH values of 2.0 and 4.0 produced slightly smaller masses of BSF larvae (0.16 g) as compared to neutral and higher pH values such as 7.0 (0.20 g) and 10.0 (0.20 g) [48]. Hence it can be said that pH plays a minor factor in BSF larvae development as compared to other more important conditions such as diet and temperature.

2.3. Small-Scale and large-scale BSF farms

For scientific research purposes, small-scale BSF farms are more ideal as they are easier to manage and regulate. Hence this allows the ease of experimentation and extraction of chemical compounds contained in BSF larvae. Small-scale BSF farms are set up as controlled environments, with the type and amount of food is known, as well as the surrounding conditions are maintained. In small-scale BSF farms, the diet can be controlled and food choice can be selected to increase certain nutritional components to one's needs. There is a debate on whether ad libitum feeding or restricted feeding is better [49], but ad libitum feeding allows the larvae to consume more food, causing the nutritional content to increase. In a small-scale environment, living space is also an important criterion to ensure BSF larvae can be nurtured successfully. Sufficient spacing helps to improve life-history traits (such as the number and size of BSF larvae) [50] and prevent large larval aggregation [51] which may lead to constant alloprophagy and autocoproprophagy in the larvae. It is also found that the supplementation of certain types of bacteria such as *Arthrobacter* AK19 in BSF larvae diets helped to enhance the growth rate and improve nutritional assimilation functions at a small-scale level [52]. These bacteria can be cultured personally on Luria nutrient agar [52] or acquired from third-party producers. Therefore, it can be said there is flexibility in designing small-scale BSF farms with many

improvements being able to be integrated into the farms. However, if looking at from a prospective business perspective, a small-scale BSF farm is less ideal as compared to a large-scale BSF farm. A small-scale farm does not have the necessary equipment and facilities to produce an output of magnitudes from large-scale farms. Large-scale farms usually have established and guaranteed markets for products and allows for easy business partnerships with other industries that thrive on the BSF rearing industry such as the animal feed industry [53]. Nonetheless, large-scale BSF farms have drawbacks, in which there needs a large capital or investment before it can be set up. A larger farm will also result in greater difficulty of quality control (which diminishes the criteria for BSF larvae research) and more environmental pollution/hygienic issues [53]. It has been noted that success in small-scale farms does not necessarily translate to the same outcome in large-scale farms. It has been found that some growth parameters from small-scale BSF farms are not scalable and cannot be replicated linearly on an industrial scale [54]. Large-scale farms also in actuality do not have proper frameworks and blueprints on how to set up a successful industrial-scale farm, so it is a particular risk venture for prospectors in this field [55]. In summary, the suitability of a small-scale BSF farm or large-scale BSF farm falls on the sole purpose and utilization of the insect. Research or extraction purposes go hand in hand with small-scale settings, while rearing and mass production are favourable for large-scale farms.

Small-scale BSF farms have been successfully set up in the past. They were done mainly for the technical basis of small-scale rearing of BSF to be used as animal feed such as fish feed [17,56]. A prototype small-scale BSF farm was designed and constructed in June 2021 for examination and extraction of vital biochemical compounds in BSF larvae in a laboratory. IoT elements were implemented into the BSF farm such as sensors and relays. The design of the small-scale BSF farm was inspired by OFERA Insects™, an aquaponic company that has attempted at constructing a compact system for rearing BSF. The IoT integration to the design had references to Sabir et al.'s BSF design system, to ensure the correct growth parameters were observed [21]. The IoT system was based on previous experiences with IoT integration of cultivating microalgae [57]. The small-scale farm was constructed as a stacked system, with three boxes being connected vertically, along with another box at the base acting as a support. The top box is a transparent plastic box, having the dimensions of 55 × 36 × 40 cm. The other boxes are blue plastic boxes, with a black lid, and have measurements of 53 × 34 × 25 cm. The stacked system has an approximate height of 1.20 m. The lower box would be the rearing area of the larval stage, where they are fed at various times throughout the day. The lower box and the middlebox are connected with two pipes measuring 55 mm in diameter, placed inclined on the opposite sides of the box (as shown in Fig. 2). This is to allow BSF larvae that are nearing the end of their larval stage and evolving into the pupal stage, to crawl upwards and seek a dry place to pupate. During this period, the larvae tend to be more mobile hence can traverse longer distances as compared to the start of their larval stage. The pipes would lead them to a drop-off into plastic containers where the BSF larvae are collected. As the main purpose of this farm is to collect and examine the BSF larvae, a large portion of the larvae will be collected, and the remaining few are allowed to pupate and continue the life cycle. This is to ensure that the life cycle is self-sustainable and allow for more generations of BSF larvae to be produced. On the other hand, the egg collection procedure was stopped when the adult BSFs have died [17]. There is a difference in methodology as the purpose of their experiment was just setting up a technical basis, whereas the goal of this experiment is to carry out experimentation on the rearing of BSF larvae for food waste reduction. The remaining pupae are allowed to undergo metamorphosis and evolve into flies. The middle box's cover is punctured with holes to allow the flies to fly to the top level. The top box is transparent to allow for easy monitoring and tracking of the adult stages of the BSF. At the top level, its



Fig. 2. PVC pipes on opposites connect the lower box to the upper box. The pipes are adhered to the sides using Liquid Nail.

purpose is to attract BSF for fertilization and oviposition of the eggs. Attractants such as sugar and water, as well as corrugated cardboard, are placed at the top level to increase the probability of success of breeding BSF eggs. Care must be taken to maintain good housekeeping and cleanliness of the top level's surroundings, as previous studies have shown that adult BSFs have great tendencies to lay their eggs directly on substrates (food), especially within areas with a spilt substrate [58]. The top box is adhered to the middle box's cover, while the middle box adheres to the lower box's cover. Hence the covers can easily be separated to observe the various levels of the farm. In the lower level, a water pipe measuring 6 mm in diameter is circled on the sides of the box, with mist nozzles installed along the way of the pipe. It is controlled by IoT components which will be discussed in a further section.

In actuality, many abiotic factors within the BSF small-scale farm can be monitored through the implementation of IoT. For example, a key factor such as temperature can be monitored through means of IoT. As mentioned above, BSF larvae are very sensitive to temperature and might produce adverse effects if the temperature gets too cold or too hot [31]. Hence temperature sensors can be installed in the larval growth area to ensure a suitable temperature for BSF larvae development is maintained. Besides that, some temperature sensors have built-in humidity sensors, hence both the temperature and moisture content can be monitored through a single sensor device. Light sensors can also be installed to ensure light intensity is at an optimum condition and not too dim or bright. Light sensors play a big role in the fertilization and oviposition of eggs [45], hence they should be installed in the breeding area of adult BSF. Water pumps can also be integrated into the IoT system, to ensure the feed or environment is sufficiently damped for the proper growth of BSF larvae.

3. Introduction to IoT, smart farming and their implementation

3.1. Introduction to IoT

Internet-of-Things (IoT) is a relatively modern concept but has been proven to be still relevant and important in this digital age. The origin of the concept of IoT has been widely disputed. Some believe that the concept of IoT and its term was first coined in September 1985 by Peter T. Lewis, a wireless technology developer during a speech to the Congressional Black Caucus Foundation 15th Annual Legislative Weekend in Washington, D.C. [59]. His first definition of the

concept of IoT is that it is the integration of people, processes, and technology with connectable devices and sensors to enable remote monitoring, status, manipulation and evaluation of trends of such devices. On the other hand, some people believed it was first introduced in 1999 by MIT's Kevin Ashton during his promotion of radio frequency identification (RFID) technology [60]. Regardless of the origin, the early definitions proved to be accurate and comparable to modern definitions available today. A version of the concept of IoT is that it is a new technology paradigm envisioned as a global network of machines and devices capable of interacting with each other [61]. In short, IoT is the interaction of "things" (i.e., sensors, relays, and software) with other technological systems through the Internet. As the popularity of IoT implementation is increasing rapidly, it is forecasted that the installed base value of 15.4 billion devices in 2015 will increase to 75.4 billion in 2025 [62]. IoT has constantly been integrated into our daily lives and many systems are utilizing its technology such as home security systems, smart farming, and connected cars.

An IoT system can be said to be composed of five major components: sensors, gateways, cloud ecosystem, data analysis, and user interface (UI) [63]. The first main component is the sensors/ "things". These smart devices are components present in what some people call the sensing layer or sensing domain [64,65]. Their main function is to collect data and information from the surroundings they are placed in and transmit the information to the succeeding components/devices. Examples are temperature and humidity sensors that can record the temperature and moisture content at any given time, motion detectors, and RFID tags. To facilitate the transfer and processing of the data acquired, these sensors and smart devices can be connected to wireless networks such as Wi-Fi, Bluetooth, ZigBee, WPAN, and many more. Every component in IoT carries a digital identity known as a universally unique identifier (UUID) [64]. This UUID can be used to identify objects on the Internet as everything has a connection.

The data collected is then passed through an IoT gateway. An IoT gateway is a physical device or virtual platform that receives the sensed data and transmits the data towards the application platform for cloud storage and data analysis [66,67]. It is also bidirectional, meaning it can work both ways by receiving information from the application platform and giving signals to the sensors or actuators to carry out the specific task. Another key function of an IoT gateway is that it pre-processes the data received (compiling and synchronizing data) before passing it to the cloud. This helps reduce the burden and load on the cloud system, hence it helps to reduce response times and network transmission costs [66]. The IoT gateway can offer security by transmitting data through higher-order encryption methodologies and protecting the system from possible cyberattacks [63,68].

At the IoT cloud, there is a massive network of servers that supports IoT components and applications. The IoT cloud offers the necessary infrastructure, storage, and servers for large-scale and real-time operations. This allows the cloud to process billions of IoT devices and traffic management at incredibly high speeds. The scenario where IoT smart devices are integrated with a cloud ecosystem is known as CloudIoT [69] or Cloud of Things (CoT) [70]. Through this integration, the advantages of cloud computing can be reaped [71]. Some of the benefits of cloud computing are reducing expendable IT costs for expensive hardware storage, the freedom and flexibility of business scalability, and allowing the ease of collaboration with other partners through access to the cloud [72]. Hence the collected data from the sensors in the BSF farm can be stored in the cloud easily.

The information and data stored in the cloud usually have to go through analytics. IoT analytics is the application of data analysis tools to analyze and interpret information collected by the sensors which can be utilized to improve and monitor the IoT system [73]. A major benefit of having an efficient IoT system with good analytical tools is that the analysis process can be carried out in real-time which helps technicians and engineers to identify the irregularities and

inconsistencies in the data quickly and prevent alarming issues. Nowadays, big corporations utilize data analytics to help identify future insights and plan out future business opportunities. In the case of the BSF farm, info regarding the temperature or humidity can be analyzed and compared with one another to see if there is a possible issue in the abiotic conditions or to identify random errors during the rearing process.

The final important aspect of an ideal IoT system should be a well-designed and well-produced user interface (UI). UIs are the visible and interactable part of the IoT system which can be accessed by common users or consumers. They allow users to visually monitor the changes in the IoT system, and can also allow the user to activate certain actuators or sensors remotely. Good UIs are usually user friendly and have a minimalistic control system to allow ease of completing tasks and prevent confusion. Typical UIs are applications on mobile phones and computers, but in recent years UIs can be present on smartwatches or virtual assistants such as Amazon Echo or Alexa [74]. If looked at commercially, UI design is crucial in today's competitive IoT market, as the visual appearance will be what draws potential purveyors to the product. UI applications need to have a clear logic flow and creative outlook to be likened by users [75]. For the BSF smart farm, UIs can be installed on applications on computers and mobile phones, which can then allow users to monitor the farm remotely through interactable widgets and controls. An example of UI monitoring will be showcased below.

3.2. Smart farming

Smart farming is a new and emerging concept that is based upon the management of farms using technologies such as IoT, robotics, and artificial intelligence (AI) to increase the output and quality of products while minimizing the manpower required for hard labor [76]. Therefore, smart farming can be defined as a management concept that provides the agricultural industry with the needed infrastructure to allow for advanced technology [77]. As smart farming comprises the principles of IoT, it also implements the usage of sensors for soil scanning and growth parameters management, telecommunication devices such as GPS, and data analytical tools along with software for IoT-based problem solving and decision making [78]. Smart farming usually encompasses a cycle that is based on IoT fundamentals. It can be divided into 4 phases: observation, diagnostics, decisions, and action [76]. The observation phase is where the digital sensors record information from the crops, livestock, or surroundings. Proceeding to the diagnostics phase, the sensorial information is passed through the gateway and fed to the IoT cloud. The decisions phase is where the user utilizes machine learning-driven components of the IoT structure to make an informed selection. Finally, the action phase is when the action is taken and the cycle is repeated. It is noted that even though the smart farming concept was created based around the agricultural industry, the ideology can be transferred to the rearing of insects as they are very similar in terms of growth needs and parameters.

Smart farming is very beneficial and provides an upper hand compared to the traditional manual farming methods. It helps to improve the quantity and quality control of the products through IoT automation. With great control over the agricultural system, it brings about better cost management and waste reduction. Hence, it can be said that gives way to a sustainable agricultural system. IoT in smart farming allows real-time monitoring which allows for producers to alter certain growth factors depending on the current environmental conditions. This enables quicker action to be taken and prevents any detrimental setbacks. Smart farming has also allowed the ease of precision farming. Precision farming is the concept of being able to get crops or livestock the precise treatment or necessity with pinpoint accuracy [76]. It relies on satellite imagery as decisions are usually made according to the area or per commodity [79].



Fig. 3. The prototype BSF small-scale farm.

Smart farming also helps to boost agricultural profitability for producers and farmers. By reducing the amount of resources needed to be inputted, money and labor costs can be reduced [80]. Through IoT, yield projections and disease/disaster probability maps can be constructed which drastically helps to reduce business risks and allow optimal cultivation of crops and animals [80]. To further elaborate on the sustainable agricultural system mentioned above, smart farming can reduce the ecological footprint. Precision farming allows the alleviation of leaching problems and greenhouse gas emissions from the excessive, unaccounted usage of fertilizers and pesticides in crop growing [81].

In smart farming, agricultural drones can be implemented to assist in the monitoring and field analysis process [82]. They can be both ground-based and aerial-based, depending on their defined function. They can carry out repetitive tasks for prolonged periods, and at top-notch efficiency. This would greatly eliminate the reliance on manual labor. They can also cover large grounds in a short amount of time, thus boosting the efficiency of the surveying and nurturing process. Besides that, smart greenhouses, which are greenhouses leveraged with IoT, help to create a self-regulating microclimate so that crops can grow in a conducive environment [83]. The real-time data received by farmers allow them to make adjustments to the system to ensure the plants have unhindered growth. Smart farming has also opened opportunities for a more efficient method in detecting and monitoring pests and diseases in crops [84]. In India, many mobile applications such as Kheti-Badi (social initiative app to present information and issues to farmers in India) and Pusa Krishi (provides simple solutions to agricultural problems faced by farmers in India) as well as Pest Management Information Systems (PMIS) have been utilized to help carry out surveillance and divulge important information to farmers [84]. This way, farmers are kept in the loop and can add technological and IoT advances to their repertoire. It may seem smart farming is only confined to the realms of agriculture, but in actuality, it has started to be integrated into the insect rearing field. Aspire Food Group has implemented IoT smart farming in rearing insects by applying complex modelling simulations and analytical approaches to optimize their insect farming techniques [85]. They are trying to build an autonomous and centralized system and

implement zero-waste systems so the farms can function at high efficiency and sustainability.

3.3. Implementation of IoT in small-scale BSF farms

As mentioned in the above sections, the prototype small-scale BSF farm created has IoT elements integrated into it as shown in Fig. 3, and they can be divided into mainly three parts: sensors/devices, software platform, and mobile application (user interface). The small white box is where the devices and firmware are stored, and the green container is the water storage for mist nozzle spraying. The blue device on the water storage is the water pump motor.

3.3.1. Sensors/Devices

The first part involves sensors, relays and moving components. As temperature and humidity conditions are vital for the successful growth of BSF, a temperature-humidity sensor should be used to monitor these factors to ensure they are maintained at an optimum range. In the prototype BSF farm, a DHT22 sensor is used. The DHT22 sensor is a 2-in-1, temperature and humidity sensor composed of a capacitive humidity sensor and a thermistor to measure the surrounding air and produce a digital signal on its data pin [86]. It is low cost and has a basic design, making it suitable for simple data logging. It has a voltage range of 3–5 V and a maximum current of 2.5 mA while requesting data [86]. Fig. 4(a) shows the DHT22 sensor attached to the wall of the lower box. Besides that, to regulate the moisture content of the rearing area, water needs to be sprayed periodically to ensure the humidity is at the right level. Water nozzles with a misting setting are connected along a water pipe surrounding the box, as shown in Fig. 4(b). The pipe is connected to a water pump that runs on a voltage of 5 V. However, the NodeMCU firmware board runs on 3 V. Hence, a relay connected to a breadboard is used to step down the voltage to the suitable value of 3 V.

3.3.2. Software platform

All the sensors require a platform to send the acquired data to, hence a NodeMCU board is used. NodeMCU is an open-source firmware and development board based on the Lua scripting language

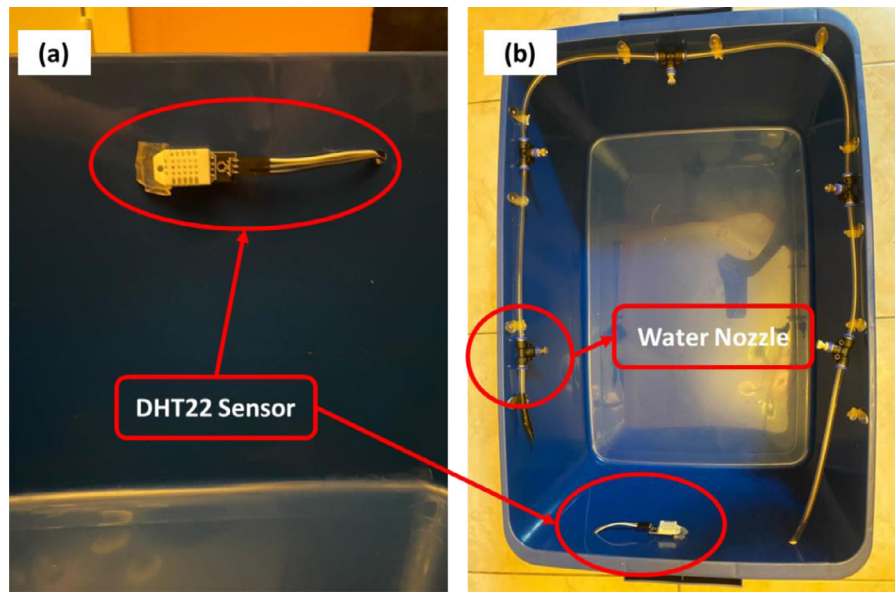


Fig. 4. (a) DHT22 sensor adhered to the wall of the rearing area (b) Waterpipe with misting nozzles surrounding the rearing area.

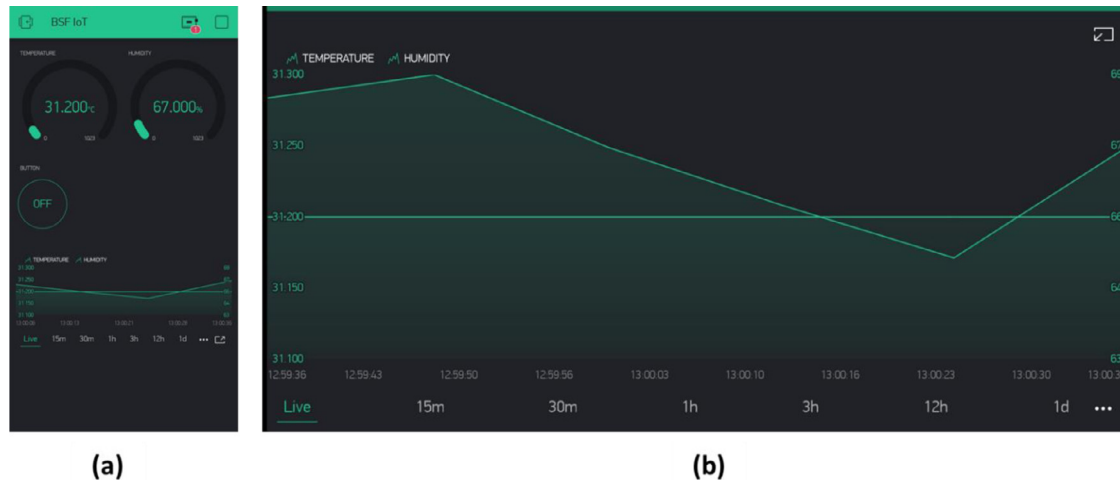


Fig. 5. (a) Blynk User Interface for the BSF Farm IoT Project (b) Superchart graph display the temperature and humidity over time.

[87]. It typically runs on a voltage of about 3 V. It can connect to the Internet with a Wi-Fi SoC (System on a Chip) called ESP8266. The NodeMCU is coded using Arduino through the C++ language to be able to receive the sensed data and transmit data and responses to a mobile application through the Wi-Fi SoC. The code requires the input of the WiFi SSID as well as the WiFi password.

3.3.3. Mobile application

The mobile application used is Blynk, an IoT platform with a free IoT Cloud network. Through this app, the sensed data can be visualized and be represented in charts and graphs. Besides that, the user interface can be customized to one's personal preference. Moving parts such as the pump can be controlled through in-app widgets at one's desire, to ensure sufficient water is provided to the rearing area of the BSF larvae. Some features of the application are shown in Fig. 5. The temperature and humidity recorded by the DHT22 sensor are displayed, as well as the trend in the graph of the respective quantities for a given set of times. The button displayed controls the motors and by activating it, water will be drawn in and sprayed through the misting nozzles. A closer look at the graph of temperature and humidity against time is shown on the right.

3.4. Related examples

In addition to the prototype BSF smart farm attempted, there have been multiple types of research done more thoroughly where IoT is implemented into the designing of smart farms, as well as BSF rearing [21,88–90]. For example, Ryu, et al. [88] has come up with a connected farm based on IoT systems, which aim to provide smart farming systems for end-users. The research used similar IoT principles and components to the prototype carried out. Mobius, an IoT service platform was used to create virtual representations of physical IoT devices [88]. The IoT service platform is compliant with the oneM2M initiative, which is an initiative started in 2012 to regulate and standardize a common machine to machine (M2M) and IoT service layer platform for an M2M service that is applicable worldwide and access-independent [89]. Besides that, a device software platform, such as a middleware known as &Cube was installed into the IoT gateways. It functions as a Java program [88]. The &Cube was installed in a Raspberry Pi, as compared to the prototype where a NodeMCU ESP8266 was used instead. As this farm was geared to agricultural purposes, more advanced sensors in addition to the temperature and humidity sensor were used, such as CO₂ sensors,

photosynthetic photon flux density (PPFD) sensors, and soil moisture sensors [88]. Instead of a Wi-Fi-based network, a ZigBee-based network was used. There are pros and cons to both. ZigBee consumes a very low amount of power and has a long battery life, however, its data transfer speed is much lower than Wi-Fi networks (maximum speed is 250kbps) [90]. Wi-Fi on the other hand has a high bandwidth and faster data transfer speed, however, the drawback is that the power consumption is greater and might be an alarming issue if the endpoints run on battery life [90]. Hence, the selection boils down to the user's preference on which factor to prioritize. Ryu et al. also developed their mobile application for remote monitoring and control over the connected farm, hence they can customize the application to one's desire.

On the other hand, in India, a BSF containment system with IoT monitoring was set up to tackle the organic waste issue which is a more environmentally friendly management system [21]. The IoT monitoring methodology done on the prototype BSF small-scale farm was heavily inspired by the research done by Sabir et al. [21] as it proved to be a simple and easy method to integrate IoT elements into a manual project. The experiment used more high-tech sensors such as MQ7 gas sensor (detects carbon dioxide), MQ135 sensor (detects ammonia, hydrogen sulfide, and other harmful gases) and MS-1100 VOC formaldehyde sensor [21]. In terms of the breeding container, used a multi-container system that was designed to utilize the BSF larvae as a composter of organic waste. It has great potential to be used on a large scale for better organic waste management, through an electrically controlled artificial system.

Finally, in Sarawak, Malaysia, a prototype of a remote monitoring system for BSF farming was set up for showcasing the possibility of implementing an IoT remote system in an environment for rural agriculture [22]. The purpose of this research was to show a possible solution to food security issues in Southeast Asia, and BSF was chosen as the subject of rearing as there are multiple parameters to be monitored [22]. Libelium Waspote Smart Agriculture platform was used as the IoT sensor node for the collection of data. A Raspberry Pi 3B is selected as the IoT Gateway device as it has a low power consumption, along with a MiFi Modem for an Internet connection [22]. A Raspberry Pi is more suitable for professional projects due to it being one of the more advanced boards, whereas the NodeMCU used for the prototype BSF farm is more for casual projects and occasions. The monitoring system also uses the XBee Pro 900HP, a wireless radio module that helps to transmit data packets to the IoT gateway [22].

4. Extraction methods of biochemical compounds from BSF

As mentioned above, the main purpose of setting up the BSF small-scale farm was to carry out extraction on the biochemical compounds contained in BSF. Some of the biochemical compounds include lipids, proteins, and chitin/chitosan. The extraction methods will be briefly discussed.

Lipids are typically the first biochemical compound to be extracted from BSF, as it is relatively the easiest compound to be extracted, and would also result in easier extraction for the remaining biochemical compounds. This can be seen in the procedures of the following experiments. The most conventional method to extract lipids from BSF larvae is to carry out Soxhlet extraction [91]. The BSF larvae can be ground and free-dried to a powder, where Soxhlet extraction with n-hexane and 2-methylloxolane (2-MeO) respectively was carried out exhaustively [92]. In the same research, a three-stage cross-current extraction was set up using the same solvents mentioned above comparatively, whereby in each stage fresh solvents were added [92]. It was found that the cumulative oil yield extracted was the same between the two conventional and industrial methods, however, it was found that 2-MeO was overall a better lipid recovery and enhanced bioactivity [92]. Another lipid extraction process involved finely ground BSF prepupae that was extracted with

petroleum ether as the solvent and recovered through solvent evaporation under a vacuum [93]. The process was found to have a yield of 87% [93]. An extraction with enzymatic methodology was also carried out, using enzymes such as *Bacillus licheniformis* protease, pepsin, papain, and pancreatin [93]. However, after centrifugation, it was found that the yield of lipids was only 10%, which is much lower than the solvent extraction methods. Hence, it was deduced that to acquire a pure lipid fraction, it is most efficient to carry out extraction with organic solvents [93].

BSF also has an attractive protein content, which has made it a popular choice as animal feed and as a meat substitute in entomophagy. A simple one-step method to extract proteins is to treat defatted BSF pellets with NaOH in a water bath, and then neutralize and place the supernatant acquired into a centrifuge for 15 min at 4000 RPM [93]. Precipitation with 10% trichloroacetic acid (TCA) solution in acetone was then carried out and proteins were then recovered. It is found that the supernatant had an 84% protein fraction recovery, and if the residual fraction was solubilized, the combined protein recovery would be 96% [93]. Another way to extract proteins from insects is to blend nitrogen-frozen insects with demineralized water and ascorbic acid [94]. The insect suspension was then filtered and centrifuged to produce pellets, the supernatant and fat fraction [94]. The protein is found in the pellets and supernatant. It is also found that defatting the insect residue beforehand and carrying out sonication enhances the protein extraction yield by a large margin [95]. Hence, it is advisable to carry out these procedures if the protein is the main subject of extraction. A study has also been done to compare chemical extraction and enzymatic-assisted extraction, and it was found that the enzymatic extraction had a greater impact on the environment as compared to the latter, due to the longer experimental time for the enzymatic process [96]. To mitigate this issue, improvements like biomass pretreatment procedures or enzymes that require a shorter extraction time should be utilized.

Chitin is also an important bioactive compound present in BSF. Chitosan, the derivative of chitin, can be produced by carrying out deacetylation on chitin, typically using hydrochloric acid and sodium hydroxide [97]. Chitin is not soluble in water, whereas chitosan is sparingly soluble in water [98]. Chitin is highly sought after, due to its versatility in many industries. It can be used as an inducer in plant defence mechanisms against diseases [99], as an additive that functions to thicken and stabilize foods and food emulsions [100], and even to produce nanofibrils, which are biopolymers that have wound-healing properties [101] and promote cell migration and proliferation [102]. Chitin is typically extracted through an initial grinding of the insect in a mortar and pestle to break the exoskeleton [103]. Then, demineralization is carried out using hydrochloric acid, followed by deproteinization with NaOH [103,104]. The solids (chitin) in the mixture are filtered out and washed with demineralized water until neutral pH is attained, and finally dried [103]. Chitin can also be extracted through biological methods, that is microbial fermentation. *Bacillus licheniformis* A6 is placed together with grounded BSF shells, and is left on a rotary shaker for 10 days [97]. The fermented powder was then washed, sieved and dried. The chitin residue then undergoes discoloration by adding sodium bicarbonate, then washed with water and dried to obtain chitin [97]. A benefit of using biological methods is that it eliminates the need of generating large volumes of acid and alkaline effluents in conventional chemical methods, which need to be treated before being able to be disposed of. Table 1 shows a comprehensive evaluation of the extraction methods discussed above.

Currently, there has not been any attempts at implementing IoT during the BSF larvae extraction process. Through IoT integration, certain extraction processes can be automated and done remotely, which will result in the reduction of manpower required on-site. IoT integration also allows monitoring of extraction parameters such as temperature and concentration, which in turn allows errors and

Table 1

Possible extraction methods for bioactive compounds in BSF.

Extraction methods	Targeted compound	Extraction conditions	Yield (%)	Potential applications	References
n-hexane Soxhlet extraction	Lipids	3-stage cross-current extraction	32.51 ± 0.39	-Biodiesel	[105,106]
2-MeO Soxhlet extraction	Lipids	3-stage cross-current extraction	35.83 ± 1.12	-Animal feed	
Petroleum ether extraction	Lipids	2-step method involving mixing and evaporation	87		
Enzymatic extraction	Lipids	Hydrolysis under varying conditions dependant on the type of enzyme used, followed by centrifugation	10		[15]
NaOH extraction	Proteins	-BSF needs to be defatted -Heated in a water bath and centrifuged at 4000 RPM for 15 mins	84 (96 if the residual protein fraction is further solubilized through acid demineralization)	-Component in processed human food -Animal feed (replacement of soy and fish meal)	
Demineralized water and ascorbic acid	Proteins	-Insects need to be frozen with N ₂ , then blended -Filtered and centrifuged	65–75 (as dry matter pellet fractions) 50–61 (as a supernatant fraction)		
Demineralization and deproteination	Chitin	-BSF is ground, then demineralized with HCl, followed by deproteination with NaOH	8–24	-Plant defence mechanism inducer	[99–102]
Microbial fermentation	Chitin	-BSF is grounded and treated with <i>Bacillus licheniformis</i> A6 -Discoloured, washed and dried	12.4	-Food additive that acts as a thickener and stabilizer -Nanofibrils (biopolymer)	

anomalies to be identified and amended accordingly. It is noted that for some extraction processes complete automation will be very difficult to achieve, however, IoT implementation will ease certain sections of the processes. Hence, there needs to be further research into the effectiveness and feasibility of implementing IoT in BSF extraction processes.

5. Future prospects and challenges in smart farming of BSF

BSF has proven to be analogous to a Swiss army knife, being versatile and applicable to many circumstances. BSF can be used as a probiotic additive and are in the works of being implemented into organic semiconductors [107]. As we are living at the apex of a technological era, it would certainly be a waste to not take the opportunity to integrate IoT into the conventional manual farming methods. Smart farming will be relevant in helping to improve and solve issues regarding manual labor in agricultural farming and insect rearing. As a majority of the world begins to have easier access to smartphones and IoT, information can easily be divulged amongst one another to provide solutions to agricultural problems faced by farmers in developing countries. These new technologies can bridge the gap between rural and urban farmers [108]. As mentioned previously, agricultural mobile applications have been developed which allows for the establishment of systems and networks for smart farming learning and information exchange.

Rearing insects on organic waste can significantly reduce waste odour and also pathogens [109,110]. Besides that, insect rearing on organic waste such as food waste and human faeces has been found to reduce organic waste in a range from 25% to 72%, on a dry matter basis [111,112]. This was done through the manual rearing process of BSF. Now with the huge boom in IoT, experiments have been carried out where IoT was implemented in the rearing process of BSF. However, as this is still a developing field in the case of BSF, there has not been any studies done in where BSF reared through IoT implementation undergo the organic waste (food waste) reduction process quantitatively, which would allow an analytical and statistical comparison between the efficiency of manually reared BSF versus IoT-reared BSF. Hence, research on this field would help to make decisions on whether one method is superior to the other.

IoT in smart farming allows for wireless technologies and remote devices to function with ease. Unmanned aerial vehicles (UAV) have

been used to survey and evaluate the growth of crops, as well as observe biodiversity and ecological landscape features [113]. With advancements in task-based control and swarm technology, drones and robots can be equipped with various devices and work together to improve land management capabilities [108]. A lot of these devices are powered through wireless networks. Wireless technologies that are suitable for long-range constraints and a large amount of data should rely on Long Term Evolution (LTE) or 4 G networks (and its future releases) to carry out technology-reliant agriculture [114]. They are the most suitable wireless networks as they can transfer large amounts of data within a short amount of time. Finally, machine learning (ML) and artificial intelligence (AI) can be implemented in smart farming to analyse and mine data for trend analysis [108]. Through ML and AI, algorithms, valuable information about productivity and growth trends can be acquired, which allows farmers to tailor to the changes needed.

However, with benefits, smart farming also brings some challenges that could be detrimental if not taken care of. Firstly, the data collected through ICT means on agriculture farming and insect rearing can raise doubts about property rights and the use of data. It is said that business models can create added value by converting the data acquired into information and advice for both farmers and regulatory authorities who have a chance of using the data for control and surveillance [80]. Hence, governments across the world need to instate rigid regulations to prevent the misuse of data, which will lead to legal and ethical issues regarding data monitoring [115].

Besides that, technology adoption to farming might pose an issue to developing countries, as there are high costs to bear and limited knowledge available to farmers from these regions [116]. Hence there is a possibility that the fruits of IoT may only be savoured by developed and industrialized countries. Hence, there needs to be an initiative whereby this technological gap in developing countries is closed and keep the farmers up to date. A social implication of IoT implementation that raises doubts in farmers themselves is whether automated and technological services will replace the existing knowledge and experience in farmers [117]. However, farmers should be able to collaborate and provide input on the trend development created by IoT devices on whether such scenarios are realistic.

Finally, as there are new technological advances, there will always be cybersecurity issues trailing behind. The increased usage of IoT in agriculture increases the level of exposure to cybercrimes, hence

causing the industry to be more vulnerable to cyberattacks on infrastructures and production facilities [118]. As the data collected usually are stored on the Internet, security and privacy issues will arise regarding smart farming threats. Cyber espionage can be rampant as countries around the world will make illegal efforts to get information and schematics regarding other nations' agricultural technology [119]. The use of GPS systems in smart farming can pose a threat by allowing cybercriminals to gain vital information regarding the farm and crops being reared [120]. This gives rise to *agroterrorism*, a term to describe cyber terrorism in the agriculture industry. Also, cloud computing attacks are frequent in IoT, as a lot of information is stored in the cloud. If a large number of vulnerable machines get infected with malware, there is a huge possibility that distributed denial-of-service (DDoS) attacks can occur and hinder the usage of the cloud [121]. DDoS attacks can cause major operational (interruption of services, new training for farm operators) and financial damages [122]. Luckily, some steps have been taken to safeguard the safety of smart farming. A lightweight device authentication solution in which session keys and public keys in Peer to Peer (P2P) network are combined to expedite encryption/decryption tasks. This helps to give rise to a safer smart farming communication system [123].

6. Conclusion

As we pivot into a new decade, food wastage is still a serious issue that has plagued all parts of the world. Even though the beginning of the COVID-19 pandemic had caused a 12% increase in household food waste generation, many initiatives have been taken by scientists to mitigate food waste problems [124]. This work serves as an attempt to showcase the potential of using IoT in BSF rearing to tackle food waste issues. This work allows for remote monitoring of a BSF farm, along with real-time monitoring of growth parameters and amendments which can be done conveniently off-site, in the case of contingencies. This work also showcases the potential of smart farming, which if implemented on BSF on a large scale, can prove to be very lucrative and efficient in terms of producing BSF larvae for various purposes, including organic waste management and also as a substance to be extracted. Besides that, this work has shown BSF larvae to be a good source of bioactive compounds with wide availability of applications in various fields. Through the implementation of IoT in these processes, the reliance on technology will help to ease the procedures of carrying out the necessary activity.

Credit Author Statement

Josiah Cheng Foong Van: Conceptualization, Josiah Cheng Foong Van: Writing – Original Draft, Josiah Cheng Foong Van, Pei En Tham, Hooi Ren Lim, Kuan Shiong Khoo, Jo-Shu Chang, Pau Loke Show: Writing – Review & Editing, Josiah Cheng, Foong Van: Visualization, Kuan Shiong Khoo, Jo-Shu Chang, Pau Loke Show: Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the Fundamental Research Grant Scheme, Malaysia [FRGS/1/2019/STG05/ UNIM/02/2], MyPAIR-PHC—Hibiscus Grant [MyPAIR/1/2020/STG05/UNIM/1] and Kurita Water and Environment Foundation (KWEF) [21Pmy004–21R]. The authors also gratefully acknowledge the financial support by Taiwan's Ministry of Science and Technology (MOST) under grant nos.

108-2218-E-029-022-MY3, 110-2221-E-029 –004 –MY3, 110-2621-M-029 –001, and 109-2622-E-110 –011.

References

- [1] Chapagain AK, James K. Chapter 12 - Accounting for the Impact of Food Waste on Water Resources and Climate Change. In: Kosseva MR, Webb C, editors. *Food Industry Wastes*. San Diego: Academic Press; 2013. p. 217–36.
- [2] Bajželj B, Richards KS, Allwood JM, Smith P, Dennis JS, Curmi E, CA Gilligan. Importance of food-demand management for climate mitigation. *Nat Clim Change* 2014;4(10):924–9.
- [3] Abeliotis K, Lasaridi K, Costarelli V, Chroni C. The implications of food waste generation on climate change: The case of Greece. *Sustain Prod Consum* 2015;3:8–14.
- [4] Jeon Y-S, Yang J-S, Park E-R, Yang J-W, Baek K. Continuous electrochemical removal of salts from Korean food wastes. *J Taiwan Inst Chem Eng* 2016;64:142–5.
- [5] Cameron M, Marshall J, Wang J, Elliot A. Solid waste in Canada. *Human Activity and the Environment, Annual Statistics*. Statistics Canada 2005.
- [6] Tejera J, Hermosilla D, Gascó A, Miranda R, Alonso V, Negro C, Blanco Á. Treatment of mature landfill leachate by electrocoagulation followed by Fenton or UVA-LED photo-Fenton processes. *J Taiwan Inst Chem Eng* 2021;119:33–44.
- [7] Nguyen TTX, Tomberlin JK, Vanlaerhoven S. Ability of Black Soldier Fly (Diptera: Stratiomyidae) Larvae to Recycle Food Waste. *Environ Entomol* 2015;44(2):406–10.
- [8] Shelomi M, Wu M-K, Chen S-M, Huang J-J, CG Burke. Microbes Associated With Black Soldier Fly (Diptera: Stratiomyidae) Degradation of Food Waste. *Environ Entomol* 2020;49(2):405–11.
- [9] Liu T, Awasthi MK, Awasthi SK, Duan Y, Zhang Z. Effects of black soldier fly larvae (Diptera: Stratiomyidae) on food waste and sewage sludge composting. *J Environ Manage* 2020;256:109967.
- [10] Craig Sheppard D, Larry Newton G, Thompson SA, Savage S. A value added manure management system using the black soldier fly. *Bioresour Technol* 1994;50(3):275–9.
- [11] Sprangers T, Noyez A, Schildermans K, De Clercq P. Cold Hardiness of the Black Soldier Fly (Diptera: Stratiomyidae). *J Econ Entomol* 2017;110(4):1501–7.
- [12] Newton L, Sheppard C, Watson D, Burtle G, Dove R. Using the Black Soldier Fly, *Hermetia illucens*, as a Value-Added Tool for the Management of Swine Manure 2005;17 Animal and Poultry Waste Management Center.
- [13] Park HH. "Black Soldier Fly Larvae Manual." https://scholarworks.umass.edu/cgi/viewcontent.cgi?article=1015&context=sustainableumass_studentshows (accessed 15 July, 2021).
- [14] Diener S, Zurbrugg C, Tockner K. Conversion of organic material by black soldier fly larvae: establishing optimal feeding rates. *Waste Manag Res* 2009;27(6):603–10.
- [15] Müller A, Wolf D, HO Gutzeit. The black soldier fly, *Hermetia illucens* – a promising source for sustainable production of proteins, lipids and bioactive substances. *Naturforsch C* 2017;72(9–10):351–63.
- [16] Mohd-Noor S, Wong C-Y, Lim J-W, Mah-Hussin M-T-A, Uemura Y, Lam M, Ramli A, Bashir M, Tham L. Optimization of self-fermented period of waste coconut endosperm destined to feed black soldier fly larvae in enhancing the lipid and protein yields. *Renew Energy* 2017;111:646–54.
- [17] Goughbedji A, Agbohessou P, Lalèye PA, Francis F, Caparros Megido R. Technical basis for the small-scale production of black soldier fly, *Hermetia illucens* (L. 1758), meal as fish feed in Benin. *J Agric Food Res* 2021;4:100153.
- [18] Bullock N, Chapin E, Evans A, Elder B, Givens M, Jeffay N, Pierce B, Robinson W. *The Black Soldier Fly How-to-Guide* 2013:12.
- [19] Wang K, Khoo KS, Leong HY, Nagarajan D, Chew KW, Ting HY, Selvarajoo A, Chang J-S, Show PL. How does the Internet of Things (IoT) help in microalgae biorefinery? *Biotechnol Adv* 2021:107819.
- [20] Einstein-Curtis A. "Black soldier fly tech aims to simplify, specialize production." <https://www.feednavigator.com/Article/2019/08/29/Black-soldier-fly-tech-aims-to-simplify-specialize-production> (accessed 3 August, 2021).
- [21] Sabir MO, Verma P, Maduri PK, and Kushagra K, "Electrically controlled artificial system for organic waste management using Black Soldier Flies with IoT monitoring," in *2nd International Conference on Advances in Computing, Communication Control and Networking (ICACCCN)*, Greater Noida, India, 2020, December 2020, pp. 871–875, doi:10.1109/ICACCCN51052.2020.9362816.
- [22] Chew KT, Jo RS, Lu M, Raman V, and Then PHH, "Organic Black Soldier Flies (BSF) Farming in Rural Area using Libelium Waspmote Smart Agriculture and Internet-of-Things Technologies," in *11th IEEE Symposium on Computer Applications & Industrial Electronics (ISCAIE)*, Penang, Malaysia, 2021, April 2021, pp. 228–232, doi:10.1109/ISCAIE51753.2021.9431801.
- [23] Ferrarezi R, Cannella L, Nassef A, and Bailey D, *UVI/AES Annual Report 2016 - Alternative Sources of Food for Aquaponics in the U.S. Virgin Islands: A Case Study with Black Soldier Flies*. 2016.
- [24] Insectta. "The BSF Life Cycle." <https://www.insectta.com/learn-more> (accessed 14 July, 2021).
- [25] Libal A. "The Difference Between Soldier Fly Larvae & Maggots." <https://animals.mom.com/difference-between-soldier-fly-larvae-maggots-8917.html> (accessed 14 July, 2021).
- [26] Dortmans B, Diener S, Verstappen B, Zurbrugg C. *Black Soldier Fly Biowaste Processing - A Step-by-Step Guide*. Eawag – Swiss Federal Institute of Aquatic

- Science and Technology Department of Sanitation, Water and Solid Waste for Development (Sandec) 2017:100.
- [27] Nakamura S, Ichiki RT, Shimoda M, Morioka S. Small-scale rearing of the black soldier fly, *Hermetia illucens* (Diptera: Stratiomyidae), in the laboratory: low-cost and year-round rearing. *Appl Entomol Zool* 2016;51(1):161–6.
 - [28] Lievens S, Poma G, Smet J, Van Campenhout L, Covaci A, Van Der Borgh M. Chemical safety of black soldier fly larvae (*Hermetia illucens*), knowledge gaps and recommendations for future research: a critical review. *J Insects as Food Feed* 2021;7:1–14.
 - [29] Smet JD, Wynants E, Cos P, Campenhout LV, HL Drake. Microbial Community Dynamics during Rearing of Black Soldier Fly Larvae (*Hermetia illucens*) and Impact on Exploitation Potential. *Appl Environ Microbiol* 2018;84(9):e02722–17.
 - [30] Shumo M, Khamis FM, Tanga CM, Fiaboe KKM, Subramanian S, Ekesi S, van Huis A, Borgemeister C. Influence of Temperature on Selected Life-History Traits of Black Soldier Fly (*Hermetia illucens*) Reared on Two Common Urban Organic Waste Streams in Kenya. *Animals* : an open access journal from MDPI 2019;9(3):79.
 - [31] Chia S, Tanga C, Khamis F, Mohamed S, Salifu D, Subramanian S, Fiaboe K, Niassy S, van Loon J, Dicke M, Ekesi S. Threshold temperatures and thermal requirements of black soldier fly *Hermetia illucens*: Implications for mass production. *PLoS One* 2018;13:e0206097.
 - [32] Tomberlin JK, Adler PH, Myers HM. Development of the Black Soldier Fly (Diptera: Stratiomyidae) in Relation to Temperature. *Environ Entomol* 2009;38(3):930–4.
 - [33] Zhang J, Huang L, He J, Tomberlin JK, Li J, Lei C, Sun M, Liu Z, Yu Z. An artificial light source influences mating and oviposition of black soldier flies, *Hermetia illucens*. *J Insect Sci* 2010;10:202.
 - [34] Zenyr Garden. "Food Amount for 1 KG of BSF Larvae & The Larvae Yield." <https://zenyrgarden.com/how-much-food-to-feed-one-kilo-of-black-soldier-fly-larvae-the-larvae-yield/> (accessed 16 July, 2021).
 - [35] Nijhout HF. The control of body size in insects. *Dev Biol* 2003;261(1):1–9.
 - [36] Simpson SJ, Sword GA, Lorch PD, ID Couzin. Cannibal crickets on a forced march for protein and salt. *Proc Natl Acad Sci U.S.A.* 2006;103(11):4152.
 - [37] Lee KP, Simpson SJ, Raubenheimer D. A comparison of nutrient regulation between solitary and gregarious phases of the specialist caterpillar, *Spodoptera exempta* (Walker). *J Insect Physiol* 2004;50(12):1171–80.
 - [38] Koelaman E. "What to feed the Black Soldier fly larvae?" <https://www.allabout-feed.net/all-about/new-proteins/what-to-feed-the-black-soldier-fly-larvae/> (accessed 16 July, 2021).
 - [39] Cammack JA, Tomberlin JK. The Impact of Diet Protein and Carbohydrate on Select Life-History Traits of The Black Soldier Fly *Hermetia illucens* (L.) (Diptera: Stratiomyidae). *Insects* 2017;8(2).
 - [40] Harnden LM, Tomberlin JK. Effects of temperature and diet on black soldier fly, *Hermetia illucens* (L.) (Diptera: Stratiomyidae), development. *Forensic Sci Int* 2016;266:109–16.
 - [41] Hopkins I, Newman LP, Gill H, Danaher J. The Influence of Food Waste Rearing Substrates on Black Soldier Fly Larvae Protein Composition: A Systematic Review. *Insects* 2021;12(7):608.
 - [42] Green TR, Popa R. Enhanced Ammonia Content in Compost Leachate Processed by Black Soldier Fly Larvae. *Appl Biochem Biotechnol* 2012;166(6):1381–7.
 - [43] Mustard JA. The buzz on caffeine in invertebrates: effects on behavior and molecular mechanisms. *Cell Mol Life Sci* 2014;71(8):1375–82.
 - [44] Tingle FC, Mitchell E, WW Copeland. The soldier fly, *Hermetia illucens* in poultry houses in north central Florida [Insect pests]. *J Ga Entomol Soc* 1975.
 - [45] Tomberlin J, Sheppard D. Factors Influencing Mating and Oviposition of Black Soldier Flies (Diptera: Stratiomyidae) in a Colony. *J Entomol Sci* 2002;37:345–52.
 - [46] Liu Z, Najjar-Rodriguez AJ, Minor MA, Hedderley DI, Morel PCH. Mating success of the black soldier fly, *Hermetia illucens* (Diptera: Stratiomyidae), under four artificial light sources. *J Photochem Photobiol B* 2020;205:111815.
 - [47] Meneguz M, Gasco L, Tomberlin JK. Impact of pH and feeding system on black soldier fly (*Hermetia illucens*, L.; Diptera: Stratiomyidae) larval development. *PLoS One* 2018;13(8):e0202591.
 - [48] Ma J, Lei Y, Ku Rehman, Yu Z, Zhang J, Li W, Li Q, Tomberlin JK, Zheng L. Dynamic Effects of Initial pH of Substrate on Biological Growth and Metamorphosis of Black Soldier Fly (Diptera: Stratiomyidae). *Environ Entomol* 2018;47(1):159–65.
 - [49] Ruhnke I, Normant C, Campbell DLM, Iqbal Z, Lee C, Hinch GN, Roberts J. Impact of on-range choice feeding with black soldier fly larvae (*Hermetia illucens*) on flock performance, egg quality, and range use of free-range laying hens. *Anim Nutr* 2018;4(4):452–60.
 - [50] Yang F, Tomberlin JK. Comparing Selected Life-History Traits of Black Soldier Fly (Diptera: Stratiomyidae) Larvae Produced in Industrial and Bench-Top-Sized Containers. *J Insect Sci* 2020;20(5).
 - [51] Barragan-Fonseca KB, Dicke M, JJA van Loon. Nutritional value of the black soldier fly (*Hermetia illucens* L.) and its suitability as animal feed – a review. *J Insects as Food Feed* 2017;3(2):105–20.
 - [52] Kooienga EM, Baugher C, Currin M, Tomberlin JK, HR Jordan. Effects of Bacterial Supplementation on Black Soldier Fly Growth and Development at Benchtop and Industrial Scale. *Front Microbiol* 2020;11(2907).
 - [53] Diener S, Lalander C, Zurbrugg C, and Vinneras B. *Opportunities and constraints for medium-scale organic waste treatment with fly larvae composting*. 2015.
 - [54] Miranda CD, Cammack JA, Tomberlin JK. Mass Production of the Black Soldier Fly, *Hermetia illucens* (L.), (Diptera: Stratiomyidae) Reared on Three Manure Types. *Animals* : an open access journal from MDPI 2020;10(7):1243.
 - [55] Green T. DipTerra LLC, editor. *Modular Farming of BSF Scaling Up - Schematic Overview* 2016:2021.
 - [56] Kenis M, Bouwassi B, Boafu H, Devic E, Han R, Koko G, Koné NG, Maciel G, Nacambo S, Pomalègn SCB, Roffeis M, Wakefield M, Zhu F, Fitches E. Small-Scale Fly Larvae Production for Animal Feed 2018:239–61.
 - [57] Tham PE, Ng YJ, Vadivelu N, Lim HR, Khoo KS, Chew KW, Show PL. Sustainable smart photobioreactor for continuous cultivation of microalgae embedded with Internet of Things. *Bioresour Technol* 2022;346:126558.
 - [58] Fisher H and Romano N. *Black soldier fly larval production in a stacked production system*. 2020.
 - [59] Sharma C. "Correcting the IoT History." <http://www.chetanasharma.com/correcting-the-iot-history/> (accessed 22 July, 2021).
 - [60] Zhang WE, Sheng QZ, Mahmood A, Tran DH, Zaib M, Hamad SA, Aljubairy A, Alhazmi AAF, Sagar S, Ma C. The 10 Research Topics in the Internet of Things. *IEEE 6th International Conference on Collaboration and Internet Computing (CIC)*, Atlanta, GA, USA, 2020; December 2020. p. 34–43. doi: 10.1109/CIC50333.2020.00015.
 - [61] Lee I, Lee K. The Internet of Things (IoT): Applications, investments, and challenges for enterprises. *Bus Horiz* 2015;58(4):431–40.
 - [62] Lucero S. IoT platforms: enabling the Internet of Things. *IHS Technology* 2016 Accessed: 22/7/2021. [Online]. Available: <https://cdn.ihs.com/www/pdf/enabling-IoT.pdf>.
 - [63] Rajiv. "What are the major components of Internet of Things." <https://www.rfpage.com/what-are-the-major-components-of-internet-of-things/> (accessed 24 July, 2021).
 - [64] Gokhale P, Bhat O, Bhat S. Introduction to IOT 2018;5:41–4.
 - [65] Hao C, Xueqin J, Heng L. A brief introduction to IoT gateway. *IET International Conference on Communication Technology and Application (ICCTA 2011)*, Beijing, China, 2011; October 2011. p. 610–3. doi: 10.1049/cp.2011.0740.
 - [66] Thales. "Bridge the Gap with IoT Gateways." <https://www.thalesgroup.com/en/markets/digital-identity-and-security/iot/inspired/iot-gateway> (accessed 24 July, 2021).
 - [67] Zhu Q, Wang R, Chen Q, Liu Y, Qin W. IOT Gateway: Bridging Wireless Sensor Networks into Internet of Things. *IEEE/IFIP International Conference on Embedded and Ubiquitous Computing*, NW Washington, DC, United States, 2010; December 2010. p. 347–52. doi: 10.1109/EUC2010.58.
 - [68] Deuskar P. "Internet of Things—a new world and a huge business opportunity." <https://www.moneycontrol.com/news/trends/expert-columns/internet-of-things-a-new-world-and-a-huge-business-opportunity-7089921.html> (accessed 24 July, 2021).
 - [69] Botta A, de Donato W, Persico V, Pescapé A. Integration of Cloud computing and Internet of Things: A survey. *Future Gener Comput Syst* 2016;56:684–700.
 - [70] Aazam M, Khan I, Alsaffar AA, Huh E. Cloud of Things: Integrating Internet of Things and cloud computing and the issues involved. In: *Proceedings of 2014 11th International Bhurban Conference on Applied Sciences & Technology (IBCAST)*, Islamabad, Pakistan, 2014; January 2014. p. 414–9. doi: 10.1109/IBCAST.2014.6778179.
 - [71] Elazhary H. Internet of Things (IoT), mobile cloud, cloudlet, mobile IoT, IoT cloud, fog, mobile edge, and edge emerging computing paradigms: Disambiguation and research directions. *J Netw Comput Appl* 2019;128:105–40.
 - [72] Queensland B. "Benefits of cloud computing." <https://www.business.qld.gov.au/running-business/it/cloud-computing/benefits> (accessed 24 July, 2021).
 - [73] Wigmore I. "IoT analytics (Internet of Things analytics)." <https://whatis.techtarget.com/definition/IoT-analytics-Internet-of-Things-analytics> (accessed 26 July, 2021).
 - [74] Leverage. "Introduction to UIs and UX in IoT." <https://www.leverage.com/iot-e-book/ui-and-ux-design-iot> (accessed 26 July, 2021).
 - [75] Maiman M. "How the User Interface on IoT Hardware Can Make or Break the User's Experience." <https://www.electronicdesign.com/technologies/iot/article/21808679/how-the-user-interface-on-iot-hardware-can-make-or-break-the-users-experience> (accessed 26 July, 2021).
 - [76] Sciforce. "Smart Farming: The Future of Agriculture." <https://www.iotforall.com/smart-farming-future-of-agriculture> (accessed 5 August, 2021).
 - [77] Bernstein C. "smart farming." <https://internetofthingsagenda.techtarget.com/definition/smart-farming> (accessed 5 August 2021).
 - [78] Zhu J, Yao Y, Li D, Gao F. Monitoring big process data of industrial plants with multiple operating modes based on Hadoop. *J Taiwan Inst Chem Eng* 2018;91:10–21.
 - [79] McBratney A, Whelan B, Ancev T, Bouma J. Future Directions of Precision Agriculture. *Precis Agric* 2005;6(1):7–23.
 - [80] Walter A, Finger R, Huber R, Buchmann N. Opinion: Smart farming is key to developing sustainable agriculture. *Proc Natl Acad Sci U.S.A.* 2017;114(24):6148.
 - [81] Schulze ED, Luyssaert S, Ciais P, Freibauer A, Janssens IA, Soussana JF, Smith P, Grace J, Levin I, Thiruchittampalam B, Heimann M, Dolman AJ, Valentini R, Bousquet P, Peylin P, Peters W, Rödenbeck C, Etiope G, Vuichard N, Wattenbach M, Nabuurs GJ, Pousi Z, Nieschulze J, Gash JH, the CarboEurope T. Importance of methane and nitrous oxide for Europe's terrestrial greenhouse-gas balance. *Nat Geosci* 2009;2(12):842–50.
 - [82] s Ahirwar, S Swarnkar, Srinivas B, Namwade G. Application of Drone in Agriculture. *Int J Curr Microbiol Appl Sci* 2019;8:2500–5.
 - [83] Meola A. "Smart Farming in 2020: How IoT sensors are creating a more efficient precision agriculture industry." <https://www.businessinsider.com/smart-farming-iot-agriculture> (accessed 5 August, 2021).
 - [84] Dutta J, Dutta J, Gogoi S. Smart farming: An opportunity for efficient monitoring and detection of pests and diseases. *J Entomol Zool Stud* 2020;8:2352–9.

- [85] Aspire Food Group. "Technology." <https://aspirefg.com/technology.aspx> (accessed 7 August, 2021).
- [86] Ada L. "DHT11, DHT22 and AM2302 Sensors." <https://learn.adafruit.com/dht> (accessed 7 August, 2021).
- [87] Components 101. "NodeMCU ESP8266." <https://components101.com/development-boards/nodemcu-esp8266-pinout-features-and-datasheet> (accessed 7 August, 2021).
- [88] Ryu M, Yun J, Miao T, Ahn I, Choi S, Kim J. Design and implementation of a connected farm for smart farming system. *IEEE SENSORS*, Busan, Korea (South), 2015; November 2015. p. 1–4. doi: 10.1109/ICSENS.2015.7370624.
- [89] Swetina J, Lu G, Jacobs P, Ennesser F, Song J. Toward a standardized common M2M service layer platform: Introduction to oneM2M. *IEEE Wirel Commun* 2014;21(3):20–6.
- [90] Ray B. "The ZigBee Vs WiFi Battle For M2M Communication." <https://www.link-labs.com/blog/zigbee-vs-wifi-802-11ah> (accessed 8 August, 2021).
- [91] Wu J, Alam MA, Pan Y, Huang D, Wang Z, Wang T. Enhanced extraction of lipids from microalgae with eco-friendly mixture of methanol and ethyl acetate for biodiesel production. *J Taiwan Inst Chem Eng* 2017;71:323–9.
- [92] Ravi HK, Vian MA, Tao Y, Degrou A, Costil J, Trespeuch C, Chemat F. Alternative solvents for lipid extraction and their effect on protein quality in black soldier fly (*Hermetia illucens*) larvae. *J Clean Prod* 2019;238:117861.
- [93] Caligiani A, Marseglia A, Leni G, Baldassarre S, Maistrello L, Dossena A, Sforza S. Composition of black soldier fly prepupae and systematic approaches for extraction and fractionation of proteins, lipids and chitin. *Food Res Int* 2018;105:812–20.
- [94] Yi L, Lakemond CMM, Sagis LMC, Eisner-Schadler V, van Huis A, van Boekel MAJS. Extraction and characterisation of protein fractions from five insect species. *Food Chem* 2013;141(4):3341–8.
- [95] Choi BD, Wong NAK, Auh J-H. Defatting and Sonication Enhances Protein Extraction from Edible Insects. *Korean J Food Sci Anim Resour* 2017;37(6):955–61.
- [96] Rosa R, Spinelli R, Neri P, Pini M, Barbi S, Montorsi M, Maistrello L, Marseglia A, Caligiani A, Ferrari AM. Life Cycle Assessment of Chemical vs Enzymatic-Assisted Extraction of Proteins from Black Soldier Fly Prepupae for the Preparation of Biomaterials for Potential Agricultural Use. *ACS Sustain Chem Eng* 2020;8(39):14752–64.
- [97] Lin Y-S, Liang S-H, Lai W-L, Lee J-X, Wang Y-P, Liu Y-T, Wang S-H, Lee M-H. Sustainable Extraction of Chitin from Spent Pupal Shell of Black Soldier Fly. *Processes* 2021;9(6):976.
- [98] Bedian L, Villalba-Rodríguez AM, Hernández-Vargas G, Parra-Saldivar R, Iqbal HMN. Bio-based materials with novel characteristics for tissue engineering applications – A review. *Int J Biol Macromol* 2017;98:837–46.
- [99] El Hadrami A, Adam LR, El Hadrami I, Daayf F. Chitosan in Plant Protection. *Mar Drugs* 2010;8(4).
- [100] Shahidi F, Arachchi JKV, Jeon Y-J. Food applications of chitin and chitosans. *Trends Food Sci Technol* 1999;10(2):37–51.
- [101] Jeffries C, Agathos SN, Rorrer G. Biogenic nanomaterials from photosynthetic microorganisms. *Curr Opin Biotechnol* 2015;33:23–31.
- [102] Jayakumar R, Prabakaran M, Nair SV, Tamura H. Novel chitin and chitosan nanofibers in biomedical applications. *Biotechnol Adv* 2010;28(1):142–50.
- [103] Soetemans L, Uyttebroek M, Bastiaens L. Characteristics of chitin extracted from black soldier fly in different life stages. *Int J Biol Macromol* 2020;165:3206–14.
- [104] Sagheer FAA, Al-Sughayer MA, Muslim S, MZ Elabee. Extraction and characterization of chitin and chitosan from marine sources in Arabian Gulf. *Carbohydr Polym* 2009;77(2):410–9.
- [105] Franco A, Scieuzo C, Salvia R, Petrone AM, Tafi E, Moretta A, Schmitt E, Falabella P. Lipids from *Hermetia illucens*, an Innovative and Sustainable Source. *Sustainability* 2021;13(18):10198.
- [106] Nguyen HC, Liang S-H, Li S-Y, Su C-H, Chien C-C, Chen Y-J, DTM Huong. Direct transesterification of black soldier fly larvae (*Hermetia illucens*) for biodiesel production. *J Taiwan Inst Chem Eng* 2018;85:165–9.
- [107] Insectta. "Biomaterials." <https://www.insectta.com/biomaterials> (accessed 27 August, 2021).
- [108] Khan N, Ray RL, Sargani GR, Ihtisham M, Khayyam M, Ismail S. Current Progress and Future Prospects of Agriculture Technology: Gateway to Sustainable Agriculture. *Sustainability* 2021;13(9):4883.
- [109] Beskin KV, Holcomb CD, Cammack JA, Crippen TL, Knap AH, Sweet ST, Tomberlin JK. Larval digestion of different manure types by the black soldier fly (Diptera: Stratiomyidae) impacts associated volatile emissions. *Waste Manage* 2018;74:213–20.
- [110] Erickson MC, Islam M, Sheppard C, Liao J, MP Doyle. Reduction of *Escherichia coli* O157:H7 and *Salmonella enterica* Serovar Enteritidis in Chicken Manure by Larvae of the Black Soldier Fly. *J Food Prot* 2004;67(4):685–90.
- [111] Surendra KC, Tomberlin JK, van Huis A, Cammack JA, Heckmann L-HL, Khanal SK. Rethinking organic wastes bioconversion: Evaluating the potential of the black soldier fly (*Hermetia illucens* (L.)) (Diptera: Stratiomyidae) (BSF). *Waste Manage* 2020;117:58–80.
- [112] Diener S, Studdt Solano NM, Roa Gutiérrez F, Zurbrugg C, Tockner K. Biological Treatment of Municipal Organic Waste using Black Soldier Fly Larvae. *Waste Biomass Valorization* 2011;2(4):357–63.
- [113] FAO. *E-Agriculture Strategy Guide: Piloted in Asia-Pacific Countries*: Food and Agriculture Organization of the United Nations, International Telecommunication Union, 2016, p. 222. [Online]. Available: <http://www.fao.org/3/i5564e/i5564e.pdf>.
- [114] Bacco M, Berton A, Ferro E, Gennaro C, Gotta A, Matteoli S, Paonessa F, Ruggeri M, Virone G, Zanella A. Smart farming: Opportunities, challenges and technology enablers. *IoT Vertical and Topical Summit on Agriculture - Tuscany (IoT Tuscany)*, Tuscany, Italy, 2018; May 2018. p. 1–6. doi: 10.1109/IOT-TUSCANY.2018.8373043.
- [115] Charo RA. Yellow lights for emerging technologies. *Science* 2015;349(6246):384.
- [116] Kutter T, Tiemann S, Siebert R, Fountas S. The role of communication and co-operation in the adoption of precision farming. *Precis Agric* 2011;12(1):2–17.
- [117] Hoeren T and Kolany-Raiser B, *Big Data in Context: Legal, Social and Technological Insights*. 2018.
- [118] Cyberterrorism Weimann G. The Sum of All Fears? *Stud Confl Terror*, 28; 2005. p. 2005129–49.
- [119] Barreto L, Amaral A. Smart Farming: Cyber Security Challenges. *International Conference on Intelligent Systems (IS)*, Funchal, Portugal, 2018; September 2018. p. 870–6. doi: 10.1109/IS.2018.8710531.
- [120] Elmaghaby AS, Losavio MM. Cyber security challenges in Smart Cities: Safety, security and privacy. *J Adv Res* 2014;5(4):491–7.
- [121] Gupta M, Abdelsalam M, Khorsandroo S, Mittal S. Security and Privacy in Smart Farming: Challenges and Opportunities. *IEEE Access* 2020;8:34564–84.
- [122] Schreier F. On Cyberwarfare. 2012;(DCAF Horizon Working Paper): pp. 1–133.
- [123] Chae C-J and Cho H-J. Enhanced secure device authentication algorithm in P2P-based smart farm system. *Peer-to-Peer Netw Appl* 2018; 11(6): 1230–1239.
- [124] Aldaco R, Hoehn D, Laso J, Margallo M, Ruiz-Salmón J, Cristobal J, Kahhat R, Villanueva-Rey P, Bala A, Batlle-Bayer L, Fullana-i-Palmer P, Irabien A, Vazquez-Rowe I. Food waste management during the COVID-19 outbreak: a holistic climate, economic and nutritional approach. *Sci Total Environ* 2020;742:140524.